ECON 470 Homework 3.1

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In [1]:
```
# Load packages

suppressPackageStartupMessages({
  library(tidyverse)
})
```

In [32]:
```
# Check data

ma.full <- read_csv("../data/output/ma-full.csv", show_col_types = FALSE)

# glimpse(ma.full)
```

Summarize the Data

Question 1

In [23]:
```
q1 <- ma.full %>%
filter(!is.na(Star_Rating)) %>%
mutate(market_share = avg_enrollment / avg_eligibles) %>%
group_by(year) %>%
summarize(
    'Mean Star Rating' = round(mean(Star_Rating, na.rm = TRUE), 2),
    'Mean Enrollment' = round(mean(avg_enrollment, na.rm = TRUE), 2),
    'Mean Market Share' = round(mean(market_share, na.rm = TRUE), 4),
    'Plan Count' = n()
    )

q1
```

A tibble: 6 × 5

year	Mean Star Rating	Mean Enrollment	Mean Market Share	Plan Count
<dbl>	<dbl>	<dbl>	<dbl>	<int>
2010	2.97	346.99	0.0125	59579
2011	3.26	388.31	0.0151	55249
2012	3.36	410.65	0.0166	58169
2013	3.55	406.73	0.0163	64174
2014	3.81	451.11	0.0179	58636
2015	3.96	482.57	0.0193	60819

Star ratings, mean enrollments, and mean market shares tend to increase over time.

Question 2

In [24]:
```
q2 <- ma.full %>%
filter(is.na(Star_Rating)) %>%
mutate(market_share = avg_enrollment / avg_eligibles) %>%
group_by(year) %>%
summarize(
    'Mean Enrollment' = round(mean(avg_enrollment, na.rm = TRUE), 2),
    'Mean Market Share' = round(mean(market_share, na.rm = TRUE), 4),
    'Plan Count' = n()
    )

q2
```

A tibble: 6 × 4

year	Mean Enrollment	Mean Market Share	Plan Count
<dbl>	<dbl>	<dbl>	<int>
2010	107.34	0.0092	48643
2011	167.90	0.0132	12754
2012	172.73	0.0119	9077
2013	213.50	0.0174	3661
2014	257.60	0.0158	3708
2015	314.21	0.0087	4680

The plans without a star rating tend to have lower average enrollments and market share.

Question 3

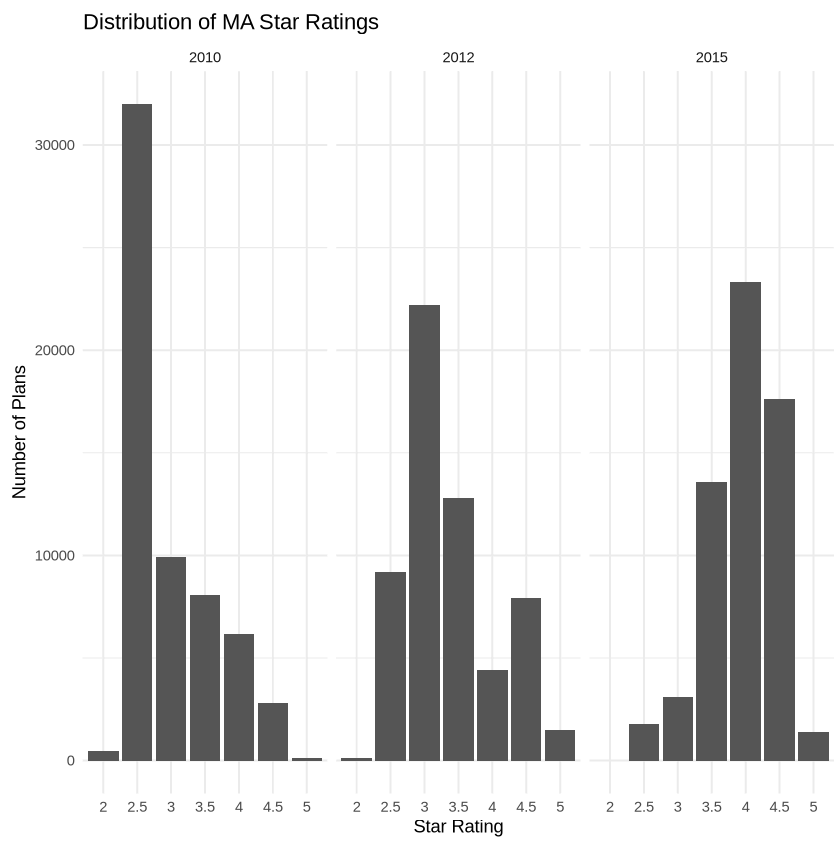
In [27]:
```
# Question 3

# Filter

stars_dist <- ma.full %>%
  filter(
    year %in% c(2010, 2012, 2015),
    !is.na(Star_Rating)
  )

# Bar graph

ggplot(stars_dist, aes(x = factor(Star_Rating))) +
  geom_bar() +
  facet_wrap(~year) +
  labs(
    title = "Distribution of MA Star Ratings",
    x = "Star Rating",
    y = "Number of Plans"
  ) +
  theme_minimal()
```



The distribution of star ratings has shifted to the right over time.

Question 4

In [31]:
```
# Indicator variables
```

```r
ma.full <- ma.full %>%
  mutate(
    market_share = avg_enrollment / avg_eligibles
  )

ma.full <- ma.full %>%
  mutate(
    star_3   = as.integer(Star_Rating == 3),
    star_35  = as.integer(Star_Rating == 3.5),
    star_4   = as.integer(Star_Rating == 4),
    star_45p = as.integer(Star_Rating >= 4.5)
  )

ma.full <- ma.full %>%
  mutate(
    star_3   = as.integer(Star_Rating == 3),
    star_35  = as.integer(Star_Rating == 3.5),
    star_4   = as.integer(Star_Rating == 4),
    star_45p = as.integer(Star_Rating >= 4.5)
  )

reg_2010 <- lm(market_share ~ star_3 + star_35 + star_4 + star_45p,
               data = filter(ma.full, year == 2010))

reg_2011 <- lm(market_share ~ star_3 + star_35 + star_4 + star_45p,
               data = filter(ma.full, year == 2011))

reg_2012 <- lm(market_share ~ star_3 + star_35 + star_4 + star_45p,
               data = filter(ma.full, year == 2012))

reg_2013 <- lm(market_share ~ star_3 + star_35 + star_4 + star_45p,
               data = filter(ma.full, year == 2013))

reg_2014 <- lm(market_share ~ star_3 + star_35 + star_4 + star_45p,
               data = filter(ma.full, year == 2014))

reg_2015 <- lm(market_share ~ star_3 + star_35 + star_4 + star_45p,
               data = filter(ma.full, year == 2015))

library(stargazer)

stargazer(reg_2010, reg_2011, reg_2012, reg_2013, reg_2014, reg_2015,
          type = "text",
          column.labels = c("2010","2011","2012","2013","2014","2015"),
          covariate.labels = c("3 Stars","3.5 Stars","4 Stars","4.5+ Stars")
          dep.var.labels = "Market Share",
          omit.stat = c("f","ser"))
```

```
=====================================================================
                              Dependent variable:
             -------------------------------------------------------
                                   Market Share
               2010      2011      2012      2013      2014      2015
               (1)       (2)       (3)       (4)       (5)       (6)
---------------------------------------------------------------------
3 Stars      0.006*** 0.008*** 0.005*** 0.011*** 0.008*** 0.005***
             (0.0004) (0.0005) (0.001)  (0.001)  (0.001)  (0.001)

3.5 Stars    0.007*** 0.009*** 0.008*** 0.009*** 0.007*** 0.006***
             (0.0004) (0.001)  (0.001)  (0.001)  (0.001)  (0.001)

4 Stars      0.008*** 0.010*** 0.004*** 0.009*** 0.007*** 0.004***
             (0.0005) (0.001)  (0.001)  (0.001)  (0.001)  (0.001)

4.5+ Stars   0.009*** 0.008*** 0.006*** 0.010*** 0.010*** 0.008***
             (0.001)  (0.001)  (0.001)  (0.001)  (0.001)  (0.001)

Constant     0.009*** 0.008*** 0.011*** 0.008*** 0.010*** 0.014***
             (0.0002) (0.0004) (0.001)  (0.001)  (0.001)  (0.001)

---------------------------------------------------------------------
Observations  18,993   19,260   20,895   23,755   22,793   22,949
R2            0.030    0.023    0.008    0.015    0.004    0.005
Adjusted R2   0.030    0.023    0.008    0.015    0.004    0.005
=====================================================================
Note:                                   *p<0.1; **p<0.05; ***p<0.01
```

Estimate ATEs

Question 5

In [29]:

```
# Question 5

data.2010 <- read.csv("../data/output/data-2010.csv")

# 1) One row per plan
data_cleaned <- data.2010 %>%
  distinct(contractid, planid, .keep_all = TRUE) %>%
  ungroup() %>%
  # 2) Calculate the running variable
  mutate(
    raw_score = rowMeans(
      as.matrix(select(., breastcancer_screen:ttyt_available)),
      na.rm = TRUE
    ),
    star_from_raw = floor(raw_score * 2 + 0.5) / 2,
    rounded_up = (star_from_raw > floor(raw_score * 2) / 2)
  )

rounded_up_table <- data_cleaned %>%
  filter(star_from_raw %in% c(3, 3.5, 4, 4.5, 5)) %>%
```

```
  group_by(star_from_raw) %>%
  summarize(
    num_plans_rounded_up = sum(rounded_up, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  complete(star_from_raw = c(3, 3.5, 4, 4.5, 5),
           fill = list(num_plans_rounded_up = 0)) %>%
  arrange(star_from_raw) %>%
  rename(Star_Rating = star_from_raw)

rounded_up_table
```

A tibble: 5 × 2

Star_Rating	num_plans_rounded_up
<dbl>	<int>
3.0	207
3.5	193
4.0	188
4.5	31
5.0	2

Question 6

In [8]:

```
# Question 6

data_cleaned <- data.2010 %>%
  distinct(contractid, planid, .keep_all = TRUE) %>%
  mutate(
    raw_score = rowMeans(select(., breastcancer_screen:ttyt_available), na.r
  ) %>%
  filter(!is.na(raw_score),
         !is.na(avg_enrollment))

estimate_rd <- function(data, cutoff, bw = 0.125) {

  rd <- data %>%
    filter(raw_score >= cutoff - bw,
           raw_score <= cutoff + bw) %>%
    mutate(
      treated = as.integer(raw_score >= cutoff),
      running_centered = raw_score - cutoff
    )

  model <- lm(avg_enrollment ~ treated + running_centered, data = rd)

  tibble(
    cutoff = cutoff,
    estimate = coef(model)["treated"],
    se = summary(model)$coefficients["treated", "Std. Error"]
```

```
  )
}

bind_rows(
  estimate_rd(data_cleaned, 3.0),
  estimate_rd(data_cleaned, 3.5)
)
```

A tibble: 2 × 3

cutoff	estimate	se
<dbl>	<dbl>	<dbl>
3.0	-2202.8193	1533.0897
3.5	118.6624	675.3635

Question 7

In [9]:

```
# Question 7

estimate_rd <- function(data, cutoff, bw) {

  rd <- data %>%
    filter(raw_score >= cutoff - bw,
           raw_score <= cutoff + bw,
           avg_enrollment > 0) %>%
    mutate(
      treated = as.integer(raw_score >= cutoff),
      running_centered = raw_score - cutoff
    )

  model <- lm(log(avg_enrollment) ~ treated + running_centered, data = rd)

  tibble(
    cutoff = cutoff,
    bandwidth = bw,
    estimate = coef(model)["treated"],
    se = summary(model)$coefficients["treated", "Std. Error"]
  )
}

bandwidths <- c(0.10, 0.12, 0.13, 0.14, 0.15)
cutoffs <- c(3.0, 3.5)

rd_results <- expand.grid(
  cutoff = cutoffs,
  bandwidth = bandwidths
) %>%
  rowwise() %>%
  do(estimate_rd(data_cleaned, .$cutoff, .$bandwidth)) %>%
  ungroup()

rd_results <- rd_results %>%
```

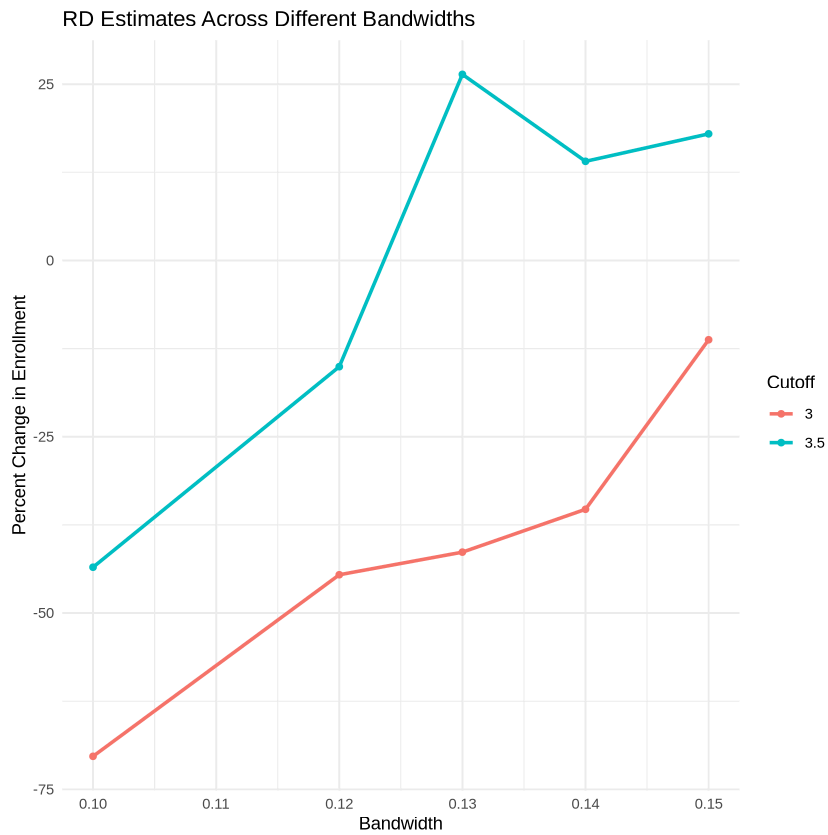
```
  mutate(percent_effect = 100 * (exp(estimate) - 1))

ggplot(rd_results, aes(x = bandwidth,
                       y = percent_effect,
                       color = factor(cutoff))) +

  geom_line(linewidth = 1) +
  geom_point() +

  labs(
    title = "RD Estimates Across Different Bandwidths",
    x = "Bandwidth",
    y = "Percent Change in Enrollment",
    color = "Cutoff"
  ) +

  theme_minimal()
```



My results are quite sensitive to the choice of bandwidth, increasing substantially as the bandwidth increases.

Question 8

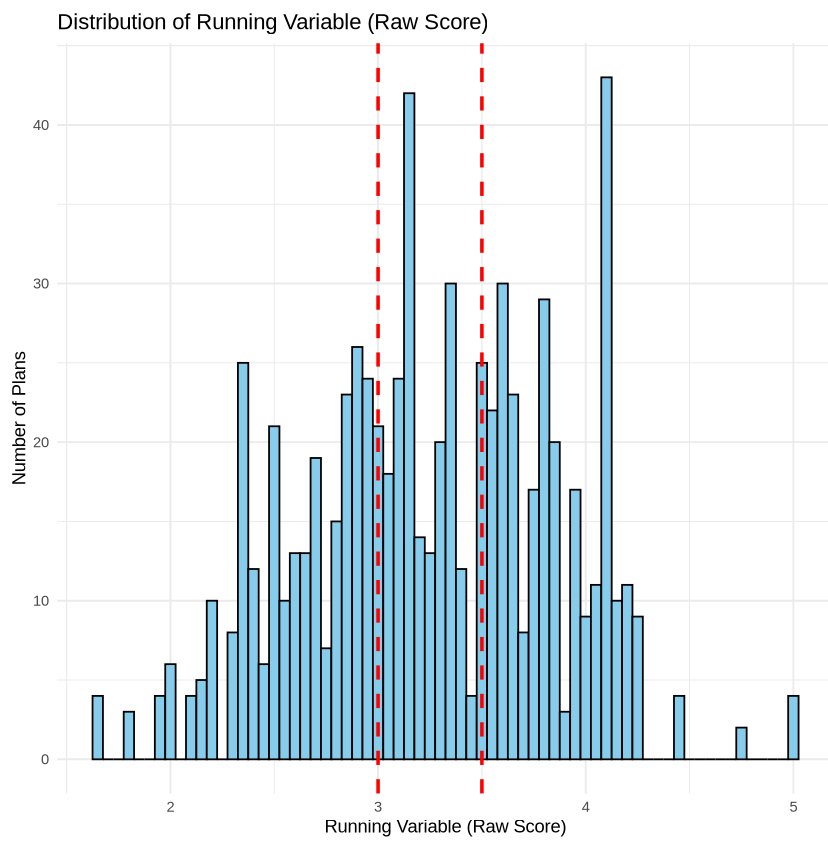
In [10]:
```
# Question 8

ggplot(data_cleaned, aes(x = raw_score)) +
  geom_histogram(binwidth = 0.05, color = "black", fill = "skyblue") +
```

```
  geom_vline(xintercept = c(3.0, 3.5),
             linetype = "dashed",
             color = "red",
             linewidth = 1) +

  labs(
    title = "Distribution of Running Variable (Raw Score)",
    x = "Running Variable (Raw Score)",
    y = "Number of Plans"
  ) +

  theme_minimal()
```



The distrubution of the running variable does not seem to change meaningfully before vs. after the 3.0 and 3.5 thresholds.

Question 9

In [11]:
```
# Question 9

data_cleaned <- data_cleaned %>%
  mutate(
    HMO = as.integer(grepl("HMO", plan_type)),
    PartD = as.integer(partd == "Yes")
  )

estimate_balance <- function(data, cutoff, bw = 0.125, outcome) {
```

```
  rd <- data %>%
    filter(raw_score >= cutoff - bw,
           raw_score <= cutoff + bw) %>%
    mutate(
      treated = as.integer(raw_score >= cutoff),
      running_centered = raw_score - cutoff
    )

  model <- lm(as.formula(paste(outcome, "~ treated + running_centered")), da

  tibble(
    cutoff = cutoff,
    outcome = outcome,
    estimate = coef(model)["treated"],
    se = summary(model)$coefficients["treated", "Std. Error"]
  )
}

balance_results <- bind_rows(
  estimate_balance(data_cleaned, 3.0, outcome = "HMO"),
  estimate_balance(data_cleaned, 3.5, outcome = "HMO"),
  estimate_balance(data_cleaned, 3.0, outcome = "PartD"),
  estimate_balance(data_cleaned, 3.5, outcome = "PartD")
)

balance_results
```

A tibble: 4 × 4

cutoff	outcome	estimate	se
<dbl>	<chr>	<dbl>	<dbl>
3.0	HMO	-0.19871579	0.1627513
3.5	HMO	-0.23686344	0.1768346
3.0	PartD	0.08526642	0.1430082
3.5	PartD	0.08854949	0.1712352

The plans before vs. after the thresholds are not statistically significantly different from one another.

Question 10

Based on my current analysis, it seems that increasing a star rating does not significantly affect MA enrollments. However, my findings were very sensitive to the choice of bandwidth, with larger bandwidths increasing the estimated treatment effect. I am still working on the code for my regression because the large negative estimate for the 3.0 plans does not seem realistic, so I plan to work on revising my code to see if the results change.